

Input window

$\tilde{\mathbf{f}}_l, \tilde{\omega}_l$ 200 Hz $[t_a - 2s, t_b + 2s]$
grey: a still phase



Anchor-frame pre-integration

$$\mathbf{R}_{a \leftarrow k} = \mathbf{R}_{Cl} \mathbf{Q}_a^T \mathbf{Q}_k \mathbf{R}_{Cl}^T, \quad \mathbf{f}_k^{C_a} = \mathbf{R}_{a \leftarrow k} \mathbf{R}_{Cl} \tilde{\mathbf{f}}_{l,k}$$

$$\hat{\mu}_f = \langle \mathbf{f}_k^{C_a} \rangle_{\text{window}}, \quad \ell_k = \mathbf{f}_k^{C_a} - \hat{\mu}_f$$

$$\Delta \mathbf{v}_k = \Sigma \ell, \quad \Delta \mathbf{d}_k = \Sigma \Sigma \ell$$

no learned parameters, 24 channels

Shared trunk

200 to 50 Hz, 0.42 M

Conv 7, stride 2

5 × dilated residual, $d = 1 \dots 16$

Conv 5, stride 2

BiGRU, 96 units

Three heads

Velocity $\hat{\mathbf{v}}_k$, upsampled to 200 Hz

Stillness $\sigma(s_k)$, BCE on reference speed

Log-variance $\eta_k \rightarrow \sigma_k$, Gaussian NLL

Structural kinematics (no free parameters except λ)

$\mathbf{v}_k \leftarrow \hat{\mathbf{v}}_k [1 - \sigma(s_k)]$ (stillness-conditioned gate)

$$\hat{\mathbf{d}}_k = \mathbf{f}_s^{-1} \sum_{i=a}^k \hat{\mathbf{v}}_i + (\mathbf{R}_{a \leftarrow k} - \mathbf{I}) \lambda, \quad \hat{\mathbf{d}}(t_a) \equiv \mathbf{0}$$

Output: one displacement stream per window, supervised at every reference frame inside it

